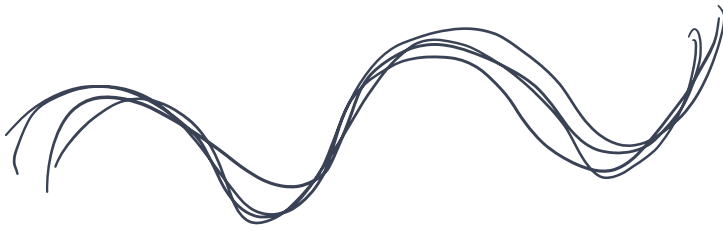


Fermi Golden Rule



$$\Gamma_{a \rightarrow b} = \frac{2\pi}{\hbar} | \langle b | H' | a \rangle |^2 \rho(E_f)$$

$$|\psi(t)\rangle = e^{-i \frac{\hat{H}}{\hbar} t} |\psi(0)\rangle$$

$$|\psi(0)\rangle = \sum_n c_n(0) |n\rangle$$

{ interaction picture }

$$\frac{\partial \psi}{\partial t} = -\frac{i}{\hbar} H \psi$$

$$i\hbar \frac{\partial \psi}{\partial t} = H \psi$$

$$\frac{d\psi}{\psi} = \frac{E}{i\hbar} dt \rightarrow \ln\left(\frac{\psi}{\psi_0}\right) = \frac{E}{i\hbar}(t-t_0)$$

$$\psi = \psi_0 e^{-i\frac{E}{\hbar}(t-t_0)}$$

$$\int_a^b \frac{dx}{x} = \ln(b) - \ln(a)$$

$$\boxed{\frac{E}{\hbar} \frac{[\cancel{\text{eV}}]}{[\cancel{\text{eV} \cdot \text{s}}]} = \frac{[\text{rad}]}{[\text{s}]} \omega}$$

$$\psi_n^{(H)} = \psi_0 e^{-i\omega_n(t-t_0)}$$

9.1 Time dependent perturbation theory

$$H = H_0 + \delta H(t)$$

no eigenstates

↓

- Hamiltonian with known wavefunction solutions

You cannot define energy eigenstates for a time dependent hamiltonian $H(t)$. WHY?

to find $|\psi(t)\rangle$ we use the interaction picture

$$\langle \psi(t) | A_S | \psi(t) \rangle$$

Schrodinger operator

$$= \langle \psi(0) | \underbrace{U^\dagger(t) A_S U(t)}_{A_H(t)} | \psi(0) \rangle$$

Heisenberg operator

$$U_0(t) = e^{-i \frac{H_0}{\hbar} t}$$

if the wavevector's evolution is simply dictated by H_0 , it can be "brought to rest" by acting upon it with the Hermitian conjugate of the time evolution operator as follows

$$U_0^\dagger(t) |\psi_0(t)\rangle = U_0^\dagger(t) U_0(t) |\psi_0(0)\rangle = |\psi_0(0)\rangle$$

but when the time evolution is governed by a time dependent Hamiltonian, the same operation, done to a general wavefunction $|\psi(t)\rangle$, acts to remove the contribution of H_0 to the time evolution.

$$|\psi(t)\rangle = U(t) |\psi(0)\rangle$$

$$= e^{\frac{i}{\hbar} (H_0 + \delta H)} |\psi(0)\rangle$$

$$(2) \quad |\varphi(t)\rangle \equiv U_0^\dagger(t) |\varphi(t)\rangle$$

$$= e^{i \frac{H_0}{\hbar} t} |\varphi(t)\rangle$$

$$e^{A+B} = e^A \cdot e^B$$

iff

$$[A, B] = 0$$

Let us write the Schrodinger equation for $|\tilde{\varphi}(t)\rangle$

$$i\hbar \frac{\partial}{\partial t} |\varphi(t)\rangle = -H_0 |\varphi(t)\rangle + e^{i \frac{H_0}{\hbar} t} \cdot i\hbar \frac{\partial |\varphi(t)\rangle}{\partial t}$$

$$i\hbar \frac{\partial}{\partial t} |\varphi(t)\rangle = [H_0 + \delta H(t)] \cdot |\varphi(t)\rangle$$

general
Schrodinger
equation

$$= [H_0 + \delta H(t)] e^{-i \frac{H_0}{\hbar} t} |\varphi(t)\rangle \quad \otimes$$

$$i\hbar \frac{\partial \varphi}{\partial t} = -\cancel{H_0} \varphi + (\cancel{H_0} + \delta H(t)) \varphi$$

$$i\hbar \frac{\partial |\varphi\rangle}{\partial t} = \delta H(t) \cdot |\varphi\rangle$$

$$(d) \quad |\varphi(t)\rangle \equiv U_0^\dagger(t) |\varphi(t)\rangle$$

$$U_0^\dagger = e^{i \frac{H_0}{\hbar} t}$$

$$\frac{\partial U_0^\dagger}{\partial t} = \frac{i H_0}{\hbar} e^{i \frac{H_0}{\hbar} t} = \frac{i H_0}{\hbar} U_0^\dagger$$

$$|\psi\rangle = U_0(t) |\varphi\rangle$$

$$i\hbar \frac{\partial |\varphi\rangle}{\partial t} = i\hbar \frac{\partial U_0^\dagger}{\partial t} |\psi\rangle + U_0^\dagger \cdot i\hbar \frac{\partial |\psi\rangle}{\partial t}$$

$$= i\hbar \cdot \frac{i H_0}{\hbar} U_0^\dagger |\psi\rangle + U_0^\dagger [H_0 + \delta H] U_0 |\varphi\rangle$$

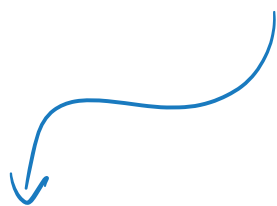
$$= -\cancel{H_0} |\varphi\rangle + U_0^\dagger \cancel{H_0} U_0 |\varphi\rangle + U_0^\dagger \delta H U_0 |\varphi\rangle$$

U_0 and H_0 commute and so

$$U_0^\dagger H_0 U_0 |\varphi\rangle$$

$$= U_0^\dagger U_0 H_0 |\varphi\rangle = H_0 |\varphi\rangle$$

$$i\hbar \frac{\partial |\varphi\rangle}{\partial t} = \underbrace{U_0^\dagger \delta H U_0}_{\tilde{\delta H}} |\varphi\rangle$$



the 0th order Heisenberg version of δH operator

$$H_0 |n\rangle = E_n |n\rangle$$

$$|\varphi(t)\rangle = \sum_{n=1}^{\infty} C_n(t) |n\rangle$$

$$|\varphi(t)\rangle = U_0^\dagger(t) |\varphi(t)\rangle$$

$$= \sum_{n=1}^{\infty} C_n(t) e^{i \frac{H_0}{\hbar} t} |n\rangle$$

$$= \sum_{n=1}^{\infty} C_n(t) e^{i \omega_n t} |n\rangle$$

expansion of the time evolved wave, function in terms of the energy basis of the unperturbed hamiltonian, with time-dependent coefficients.

$$i\hbar \frac{\partial |\psi(t)\rangle}{\partial t} = \overbrace{(H_0 + \delta H)}^H |\psi(t)\rangle$$

$$H_0 |n\rangle = \epsilon_n |n\rangle$$

$$|\psi(t)\rangle = \sum_n c_n(t) \underbrace{e^{-i\frac{\epsilon_n}{\hbar}t}}_{U_0 |n\rangle} |n\rangle$$

$$\sum_n i\hbar \frac{\partial c_n}{\partial t} e^{-i\frac{\epsilon_n}{\hbar}t} |n\rangle + \sum_n \cancel{c_n(t) \epsilon_n} e^{-i\frac{\epsilon_n}{\hbar}t} |n\rangle$$

$$= \sum_n c_n(t) \cancel{[H_0 + \delta H]} e^{-i\epsilon_n t / \hbar} |n\rangle$$

$$\Rightarrow \sum_n \left[i\hbar \frac{\partial c_n}{\partial t} - c_n \cdot \delta H \right] e^{-i\frac{\epsilon_n}{\hbar}t} |n\rangle = 0$$

$$\downarrow$$

$$\langle m | \quad " \quad \rangle$$

$$= \sum_n \left[i\hbar \frac{\partial C_n}{\partial t} \langle m|n \rangle - C_n \langle m|\delta H|n \rangle \right] e^{-i \frac{\epsilon_n}{\hbar} t} = 0$$

$$= i\hbar \frac{\partial C_m}{\partial t} e^{-i \frac{\epsilon_m}{\hbar} t} - \sum_n C_n e^{-i \frac{\epsilon_n}{\hbar} t} \delta H_{nm} = 0$$

$$= i\hbar \frac{\partial C_m}{\partial t} = \sum_n C_n \cdot \delta H_{nm} e^{i \frac{\epsilon_m - \epsilon_n}{\hbar} t}$$

$$H_0 = \begin{pmatrix} \epsilon_a & 0 \\ 0 & \epsilon_b \end{pmatrix}$$

two state system $H_0|a\rangle = \epsilon_a|a\rangle$

$$H_0|b\rangle = \epsilon_b|b\rangle$$

$$H_0 = \begin{pmatrix} \epsilon_a & 0 \\ 0 & \epsilon_b \end{pmatrix} \quad \delta H(t) = U \delta(t)$$

$$U = \begin{pmatrix} U_{aa} & U_{ab} \\ U_{ba} & U_{bb} \end{pmatrix} = \begin{pmatrix} 0 & \alpha \\ \alpha^* & 0 \end{pmatrix}$$

$$P(\psi(t>0)=|b\rangle \mid \psi(t<0)=|a\rangle)$$

$$i\hbar \frac{\partial C_m}{\partial t} = \sum_n C_n \cdot \delta H_{nm} e^{i \frac{\epsilon_m - \epsilon_n}{\hbar} t}$$

$$i\hbar \frac{\partial C_a}{\partial t} = \sum_{k=a,b} C_k \cdot \delta H_{ak} e^{i\omega_{ak} t}$$

$$= C_b \delta H_{ab} e^{i\omega_0 t}$$

$$i\hbar \frac{\partial C_b}{\partial t} = C_a \delta H_{ba} e^{i\omega_0 t}$$

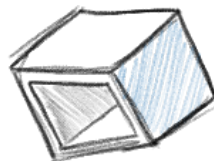
$$V^2 \cdot \frac{4 \sin^2\left(\frac{\omega t_0}{2}\right)}{(E_f - E_i)^2}$$

$$V^2 \cdot \frac{\sin^2\left(\frac{\omega t_0}{2}\right)}{\left(\frac{E_f - E_i}{2\hbar}\right)^2 \cdot \hbar^2}$$

$$V^2 \cdot \frac{\sin^2\left(\frac{\omega t_0}{2}\right)}{\left(\frac{\omega t_0}{2}\right)^2 \cdot \frac{\hbar^2}{t_0^2}}$$

$$\left(\frac{\gamma t_0}{\hbar}\right)^2 \cdot \text{sinc}^2\left(\frac{\omega t_0}{2}\right)$$

malfunction





μ

$$F = -\frac{G\mu m}{r^2}$$

ρdr
↑

$$\left[\alpha\right] = \frac{m}{r^2} \quad dF = \frac{Gm}{r^2} \left[\frac{\mu}{r} dr - d\mu \right]$$

$$\frac{\alpha}{r \rightarrow r_0} = \frac{Gm}{r^2} \left[\frac{\mu(r)}{r} dr - 4\pi \rho(r) r^2 dr \right]$$

$$m \frac{dr^2}{dt^2} = \frac{Gm}{r^2} \left[\frac{\mu(r)}{r} - 4\pi \alpha \frac{r^2}{r \rightarrow r_0} \right] dr$$

ma

$m da$

$$\omega_p^2 = \frac{e^2 n_e}{\epsilon_0 m_e^*}$$

$$\frac{m_e^* \epsilon_0}{e^2 n_e} \cdot \frac{4 \pi e^2}{m_e^*} \frac{\epsilon''(\omega_2)}{\hbar \hbar}$$

$$\hbar^2 \frac{1}{\omega_p^2} \cdot \frac{4 \pi e^2}{3} \frac{\epsilon''(\omega_2)}{m_e^*}$$

$$\epsilon_m(\omega) = \epsilon_\infty - \frac{\omega_p^2}{\omega^2 + i\omega\gamma}$$

$$\epsilon_\infty - \frac{\omega_p^2 (\omega^2 - i\omega\gamma)}{\omega^4 + \omega^2 \gamma^2}$$

$$= \epsilon_\infty - \frac{\omega_p^2 \omega^2}{\omega^4 + \omega^2 \gamma^2} + i \frac{\omega \gamma \omega_p^2}{\omega^4 + \gamma^2 \omega^2}$$

$$\epsilon''_m(\omega) = \frac{\gamma \cdot \omega_p^2}{\omega \cdot (\omega^2 + \gamma^2)}$$

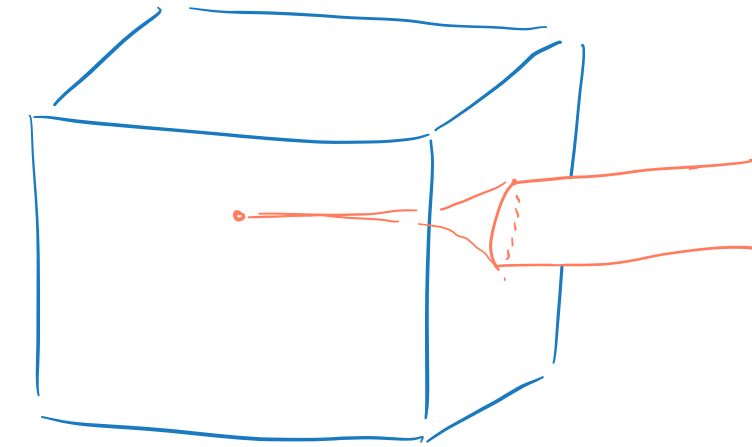
$$\mu_e = \frac{\partial U}{\partial N_e}$$

$$= \frac{\partial \left(N_e \frac{\hbar^2 K^2}{2m_e} \right)}{\partial N_e}$$

$$U = \int_0^{\infty} f(\epsilon) g(\epsilon) \epsilon \, d\epsilon$$

$$= \int_0^{\infty} \frac{f(\epsilon)}{e^{\beta(\epsilon - \mu)} + 1} \, d\epsilon$$

$$\mu = \frac{\partial U}{\partial N} = \int_0^{\infty} \frac{\partial}{\partial N} \left[- \frac{g(\epsilon)}{e^{\beta(\epsilon - \mu)} + 1} \right] d\epsilon$$



$$\frac{1}{n_e} = \frac{V}{N_e} = \frac{N \left(\frac{4\pi}{3} r_e^3 \right)}{N}$$

Volume
per
conduction
electron

$$r_e = \left(\frac{3}{4\pi n_e} \right)^{\frac{1}{3}}$$

$$\mu(T) = \epsilon_F \left[1 - \frac{\pi^2}{12} \left(\frac{T}{T_F} \right)^2 \right]$$

$$f(\epsilon; T) = \frac{1}{e^{\frac{\epsilon - \mu(T)}{k_B T}} + 1}$$

$$\frac{E_c}{E_{oc}}$$

$$E_{ov}$$

$$E_{ov} = -1.51$$

$$E_v < E_{ov}$$

$$\tilde{\epsilon}_{oc} = \epsilon_{oc} - \epsilon_F$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = E_v - E_{ov}$$

$$\tilde{\epsilon}_0 = \epsilon_0 - \epsilon_F$$

$$\frac{x^2}{c^2} - \frac{z^2}{d^2} = E_c - E_{oc}$$

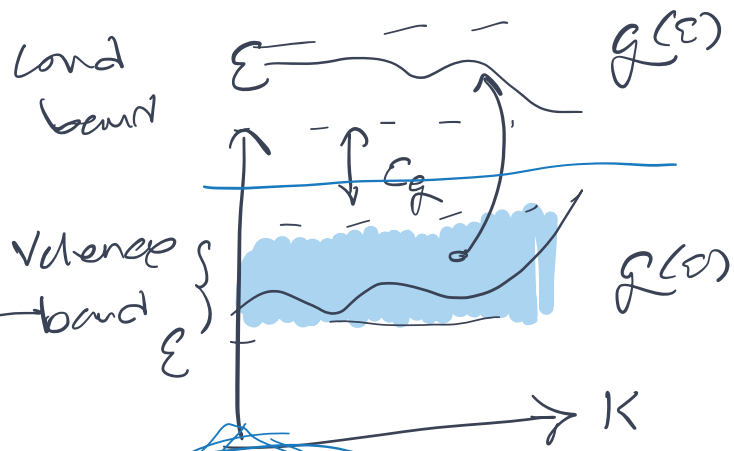
$$E_{oc} = 1.474$$

$$E_c > E_{oc}$$

$$\tilde{\epsilon}_{ov} = \epsilon_{ov} - \epsilon_F$$

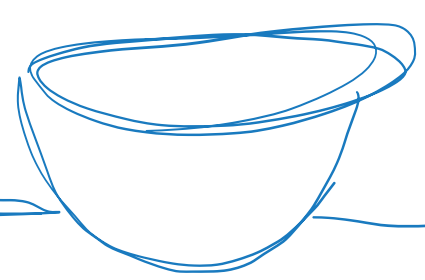
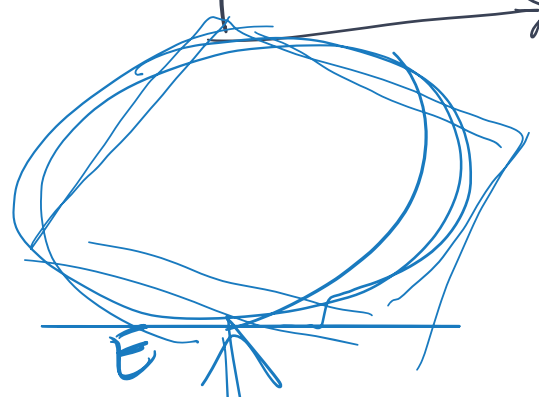
$$k_x^2 + k_y^2 + k_z^2 = \frac{2mE}{\hbar^2}$$

$$\epsilon_{ov} = \epsilon_F - \tilde{\epsilon}_{ov}$$

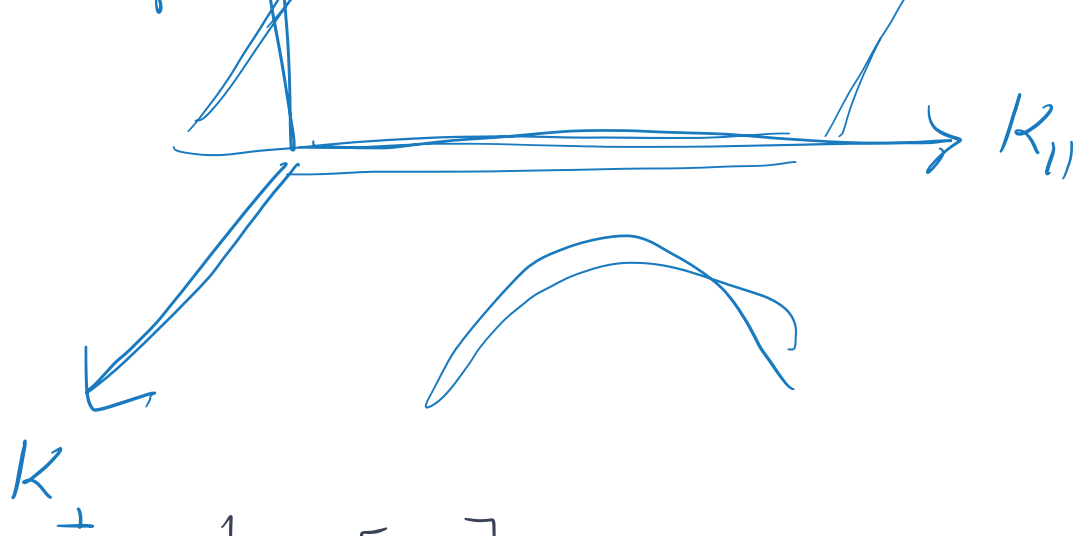


$$\epsilon_{ov} = \tilde{\epsilon}_{ov} - \epsilon_F$$

$$\tilde{\epsilon}_{ov} =$$



$$E_F$$



$k_{\perp}$

$$\left[\frac{1}{\epsilon^3} \right]$$

$$\left[\frac{i}{\epsilon^3} \right]$$

$$[\epsilon] \quad [\epsilon]$$

$$d\epsilon_1 d\epsilon_2$$

$$g(\epsilon_1) g(\epsilon_2)$$

$$d\epsilon_1 = \epsilon_{\perp} \delta(\epsilon_1 - \epsilon_2)$$

$$\bar{f}(\epsilon_2)$$

$$\int f(\epsilon_1) g(\epsilon_1) [1 - f(\epsilon_2)] g(\epsilon_2) d\epsilon_1 d\epsilon_2$$

$$f(\vec{k}_1) [1 - f(\vec{k}_2)] d^3 k_1 d^3 k_2$$

$$(2/\pi)^2 (k_1 \cdot k_2)^2 dk_1 dk_2$$

$$\iint f_v(\vec{k}_1) \bar{f}_c(\vec{k}_2) \delta(\vec{k}_2 - \vec{k}_1 - \vec{k}_{ph}) d^3k_1 d^3k_2$$

$$\hbar \vec{k}_1 + \hbar \vec{k}_{ph} = \hbar \vec{k}_2$$

$$\vec{k}_2 - \vec{k}_1 - \vec{k}_{ph} = 0$$

$$\int f_v(\vec{k}_1) [1 - f_c(\vec{k}_1 + \vec{k}_{ph})] d^3k_1$$

$$\int f_v(k)$$

$$200 \times 10^{-6} \text{ J} \left[\frac{\text{J}}{\text{pulse}} \right]$$

Needed

$$2260 \text{ KJ}$$

$$2260 \times 10^3 \text{ J}$$

$$f_{\text{pulse}} = 10 \text{ Hz} = \frac{10 \text{ pulses}}{\text{sec}} \quad \boxed{2.260 \times 10^5 \text{ KJ}}$$

$$P = \frac{200 \times 10^{-6} \text{ J}}{\text{sec}} = 2 \times 10^{-3} \text{ W} = 2 \times 10^{-6} \text{ KW}$$

$$P \cdot t = E_{\text{needed}}$$

$$t_{\text{needed}} = \frac{E_{\text{needed}}}{P} = \frac{2.260 \times 10^5 \text{ KJ}}{2 \times 10^{-6} \text{ KW}} \cong 1 \times 10^{11} \text{ sec}$$

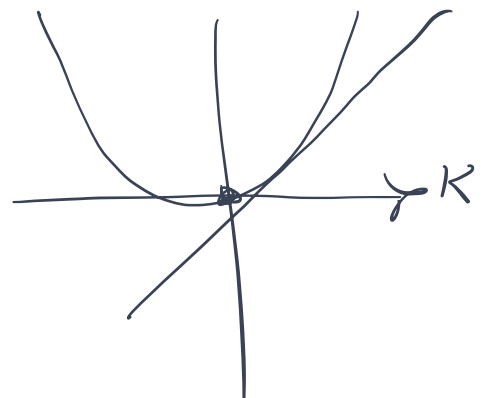
$$\boxed{Q = C_v \Delta T}$$

$$t_n = 10^{11} \text{ sec}$$

$$K = \frac{2m_e c}{\hbar}$$

$$\omega = Kc$$

$$\omega = \frac{\hbar K^2}{2m_e}$$



$$f^T(k, \omega) = \frac{1}{\exp\left(\frac{\epsilon(k, \omega) - \mu}{k_B T}\right) + 1}$$

$$\begin{aligned}\epsilon(k, \omega) &= \epsilon_e(k) + \epsilon_{ph}(\omega) \\ &= \frac{\hbar^2 k^2}{2m} + \hbar\omega\end{aligned}$$

$$I_e^T(\omega) = \int_0^\infty f^T(k, \omega) [1 - f^T(k, 0)] 4\pi k^2 dk$$

$$f(k, \omega; \omega_2) = f^T(k, \omega) + \delta E(\omega_2) \cdot B(k, \omega; \omega_2)$$

$$\begin{aligned}B(k, \omega; \omega_2) &= f^T(k, \omega - \omega_2) [1 - f^T(k, \omega)] \\ &\quad - f(k, \omega) [1 - f^T(k, \omega + \omega_2)]\end{aligned}$$

$$I_e(\omega; \omega_L) = \int_0^\infty f(k, \omega; \omega_L) [1 - f(k, 0; \omega_L)] 4\pi k^2 dk$$

$$f(\varepsilon(k) + \hbar\omega; \omega_L) [1 - f(\varepsilon(k); \omega_L)]$$

$$\delta E = \left| \frac{E_L}{E_{sat}} \right|^2 = \tau_{ee} |E_L|^2 \cdot R(\omega_L)$$

$$= \tau_{ee} |E_L|^2 \cdot \left(\frac{\Sigma_F}{\hbar\omega_L} \right) \cdot \frac{2\varepsilon_0}{3\hbar n_e} \Sigma''(\omega_L)$$

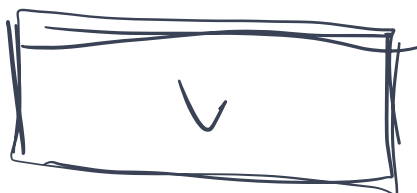
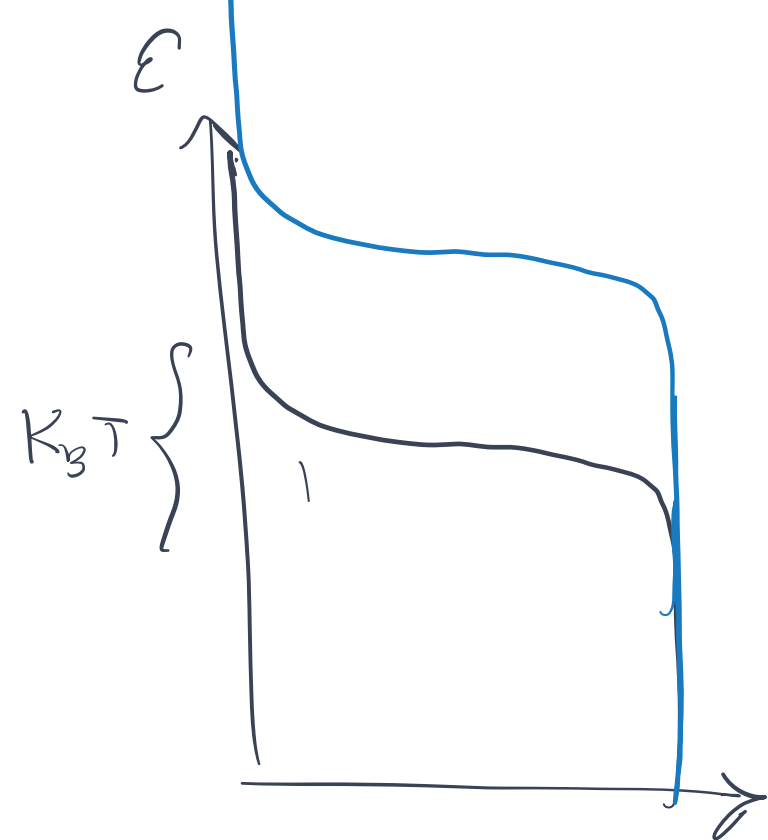
Drude
for intraband

$$\epsilon''(\omega_2)$$

$$\frac{df}{dt} = \left(\frac{\partial f}{\partial t} \right)_{e-e} + \left(-\Gamma \right)_{abs}$$

$$+ \left(-\Gamma \right)_{\text{spontaneous emission}}$$

$$\epsilon_T = k_B T$$



$$g[n] = (f * h)[n] = \sum_{m=0}^3 f[m] \cdot h[n-m]$$

$$g[0] = \sum_{m=0}^3 f[m] \cdot h[0-m]$$

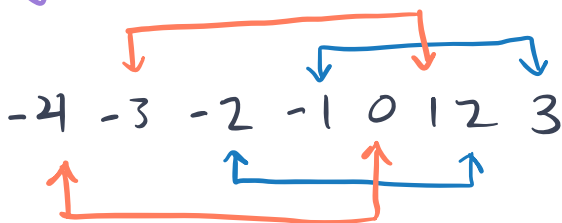
$$= f[0] \cdot h[0] + f[1] \cdot h[-1] + f[2] \cdot h[-2] + f[3] \cdot h[-3]$$

$$\otimes = f[0]h[0] + f[1]h[3] + f[2]h[2] + f[3]h[1]$$

$$h = [1, -1, 0, 0] \quad |100$$

$$= f[0] \cdot 1 + \cancel{f[1] \cdot 0} + f[2] \cdot 0 + f[3] \cdot (-1)$$

$$\Rightarrow \begin{bmatrix} g[0] \\ g[1] \\ g[2] \\ g[3] \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} f[0] \\ f[1] \\ f[2] \\ f[3] \end{bmatrix}$$



$$h[-1] = h[3]$$

$$h[-2] = h[2]$$

$$h[-3] = h[1]$$

h symmetric

$$|\langle i | V | f \rangle|^2 \delta(\epsilon_f - \epsilon_i - \hbar\omega)$$

$$\Gamma = \frac{2\pi}{\hbar} \left| \mu_{i \rightarrow f}(\epsilon_i, \epsilon_f) \right|^2 \delta(\epsilon_f - \epsilon_i - \hbar\omega)$$



single atom



statistical

$$n = \frac{N}{V} = \int \frac{f(\epsilon(\vec{k}))}{4\pi^3} d\vec{k}$$

Learning and memory

$$n = \frac{N}{V} = \int \frac{f(\epsilon(\vec{k}))}{4\pi^3} d\vec{k}$$

$$\frac{\hbar^2 k_f^2}{2m_e} - \frac{\hbar^2 k_i^2}{2m_e} = \hbar\omega$$

$$k_f^2 - k_i^2 = \frac{2m_e\omega}{\hbar}$$

$$\vec{k}_f - \vec{k}_i = \vec{q} \quad \epsilon = \hbar\omega = \hbar kc$$

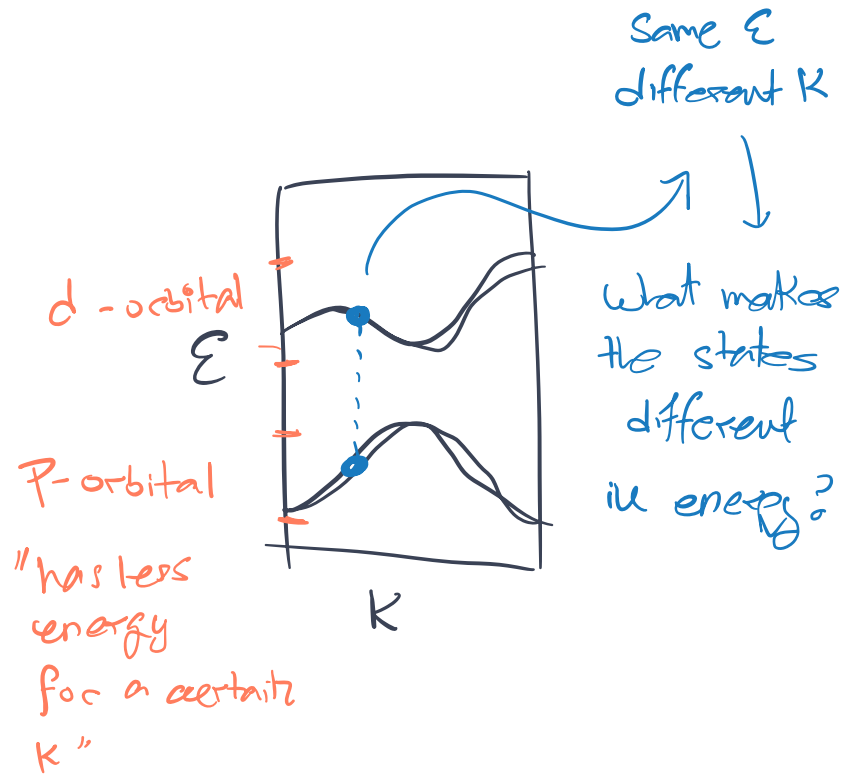
$$k = \frac{\epsilon}{\hbar c}$$

$$k_e = \sqrt{\frac{2m_e\epsilon}{\hbar^2}} \sim \sqrt{\epsilon}$$

$$k_{ph} = \frac{\epsilon}{\hbar c}$$

closed

$$|k_e^f| - |k_e^i| = \frac{\epsilon}{\hbar c}$$



it is said that to calculate spontaneous emission (which is said to be the dominant contributor to BB-radiation) one has to couple the electronic states to the EM vacuum fluctuations, which we don't do. We seemingly calculate stimulated absorption and consider it to be emission?

δ_z

$$W(\vec{k}_i) = \frac{2\pi}{\hbar} \sum_{\vec{k}_f} |\mu_{fi}|^2 \delta(\epsilon_c(\vec{k}_f) - \epsilon_v(\vec{k}_i) - \hbar\omega)$$

$$|\mu_{fi}|^2 = \left| \langle \psi_{c,\vec{k}_f} | e^{i\vec{k}_{ph} \cdot \vec{r}} \underbrace{\vec{p} \cdot \hat{n}}_{\text{photon polarization}} | \psi_{v,\vec{k}_i} \rangle \right|^2$$

$$= \underbrace{|\vec{p} \cdot \hat{n}|^2}_{\mu} \delta(\vec{k}_f - \vec{k}_i - \hbar\omega)$$

transition rate
of a single
electron state

$$\downarrow W(\vec{k}_i) = \frac{2\pi}{\hbar} |\mu|^2 \sum_{\vec{k}_f} \delta(\vec{k}_f - \vec{k}_i - \hbar\omega) \cdot \delta(\epsilon_c(\vec{k}_f) - \epsilon_v(\vec{k}_i))$$

for initial
state $\vec{k}_i$

$$= \frac{2\pi}{\hbar} |\mu|^2 \delta(\epsilon_c(\vec{k}_i + \hbar\vec{k}_{ph}) - \epsilon_v(\vec{k}_i) - \hbar\omega)$$

$$\Gamma = \frac{2}{V} \sum_{\vec{k}_i} W(\vec{k}_i) \cdot f_v(\vec{k}_i) [1 - f_c(\vec{k}_i + \vec{k}_{ph})]$$

$$\leadsto \frac{2}{V} \int \frac{d^3k}{(2\pi)^3} W(\vec{k}_i) \cdot f_v(\vec{k}_i) [1 - f_c(\vec{k}_i + \vec{k}_{ph})]$$

FBZ

counting several transitions weighed by the occupation
at $\vec{k}_i$ in V and vacancy at $\vec{k}_f$ in C .

$$\Gamma \sim \int w(k) f_v(k_i) \bar{f}_c(\vec{k}_i + \vec{k}_{ph})$$

$$= \frac{|\mu|^2}{2\pi^2 V} \int_{FBZ} d^3K f_v(\vec{K}_i) \left[1 - f_c(\vec{K}_i + \vec{K}_{ph}) \right] \cdot \delta(\epsilon_c(\vec{K}_i + \vec{K}_{ph}) - \epsilon_v(\vec{K}_i) - \hbar\omega)$$

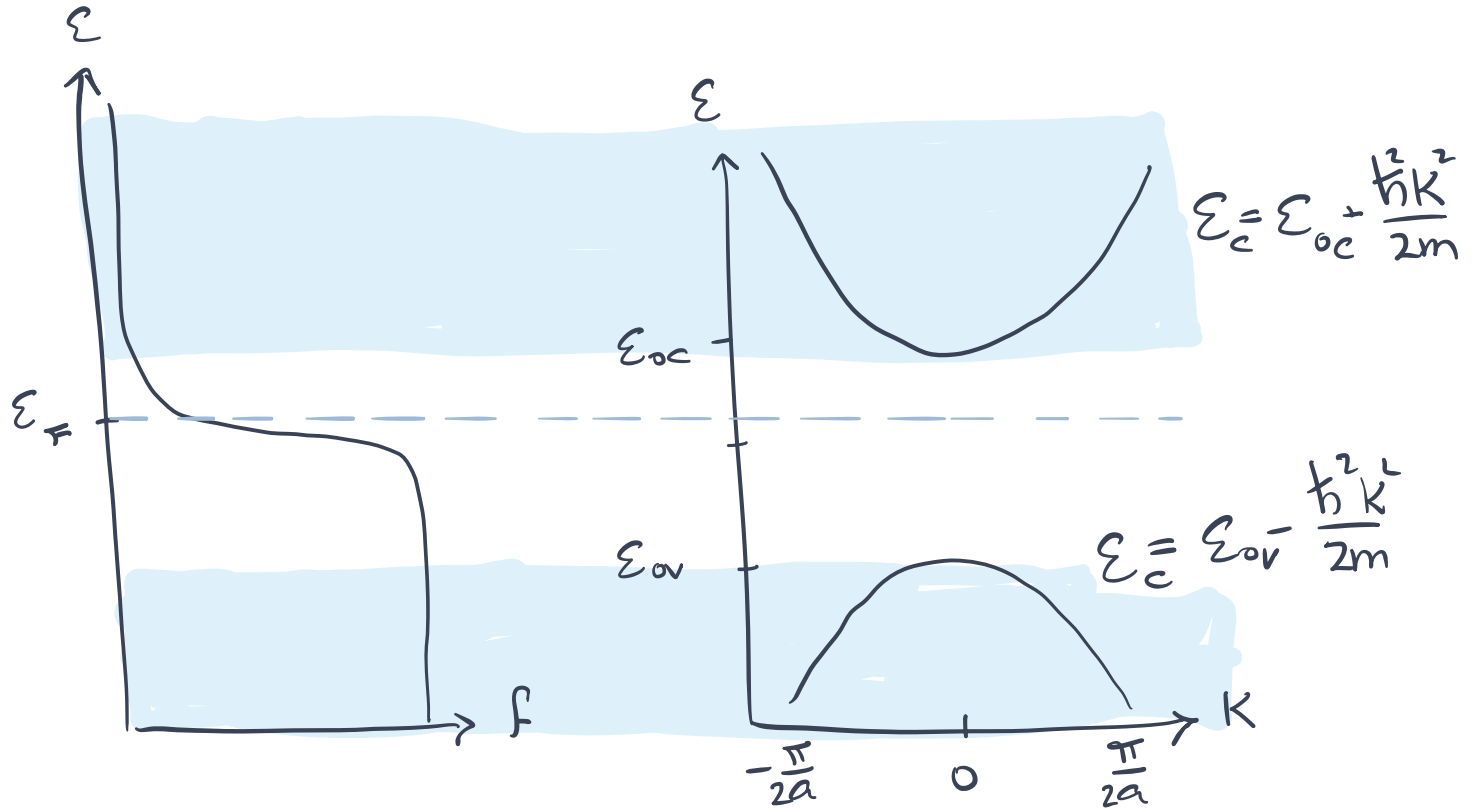
$$= \frac{|\mu|^2}{2\pi^2 V} \int_{FBZ} d^3K$$

$$\langle w \rangle = \int_{FBZ} \underbrace{\left[\int_{FBZ} w_{fi} f_v(\vec{K}_i) \frac{d^3K_i}{(2\pi)^3} \right]}_{\langle w_f \rangle} \left(1 - f_c(\vec{K}_f) \right) \frac{d^3K_f}{(2\pi)^3}$$

$\langle w_f \rangle$

$$w_{fi}(\omega) = \frac{2\pi}{\hbar} |\mu_{fi}|^2 \delta_k \delta \varepsilon$$

↓
expectation
of the rate
of transitions
from all possible i)
states to a certain f) state



$$f_v = \frac{1}{e^{\beta(\varepsilon_v - \mu)} + 1}$$

$$f_c = \frac{1}{e^{\beta(\varepsilon_c - \mu)} + 1}$$

$$\int f(\vec{k}_i) \bar{f}(\vec{k}_f) \delta(\vec{k}_f - \vec{k}_i - \vec{q}) d\vec{k}_f$$

$$(1) \quad \vec{k}'_e = \vec{k}_e - \vec{k}_{ph}$$

$$(2) \quad \mathcal{E}'_e = \frac{\hbar^2 k_e'^2}{2m_e} = \frac{\hbar^2}{2m_e} (\vec{k}_e - \vec{k}_{ph})^2$$

$$= \frac{\hbar^2 k_e^2}{2m_e} + \frac{\hbar^2 k_{ph}^2}{2m_e} + \frac{\hbar^2 \vec{k}_{ph} \cdot \vec{k}_e}{m_e}$$

$\underbrace{\hspace{10em}}_{\neq \hbar\omega}$

$$E_i = \hbar\omega + \frac{\hbar^2 k^2}{2m_e}$$

$$\vec{P}_i = \hbar\vec{k} + \hbar\vec{q}$$

$$\vec{P}_f = \hbar\vec{k}'$$

$$E_f = \frac{\hbar^2 k'^2}{2m_e}$$

$$\begin{aligned}
 E_f &= \frac{\hbar^2}{2m_e} (\vec{k} - \vec{q})^2 \\
 &= \frac{\hbar^2}{2m_e} (k^2 + \vec{k} \cdot \vec{q} + q^2) \\
 &= \frac{\hbar^2 k^2}{2m_e} + \underbrace{\frac{\hbar^2}{2m_e} (\vec{k} \cdot \vec{q} + q^2)}_{\neq E_i}
 \end{aligned}$$

intraband transitions do not conserve momentum?
or energy?

$$\alpha = \frac{\hbar^2}{2m} \quad \beta = \hbar c \quad = 0$$

$$\alpha q^2 - \beta q + Kq \cos \theta = 0$$

$$\alpha q^2 + q(K \cos \theta - \beta) = 0$$

$$q(\alpha q + K \cos \theta - \beta) = 0$$

$$q = \frac{\beta - K \cos \theta}{\alpha} = \frac{\hbar c - K \cos \theta}{\frac{\hbar^2}{2m}}$$

$$\frac{1}{\hbar^2}$$

$$\left(K_f \frac{L^2}{\hbar^2} s \right) \rightarrow K_f \cdot L$$

$$q = \frac{2m_e c}{\hbar} - \frac{2m_e K \cos \theta}{\hbar^2}$$

$$\frac{K}{\hbar} = \frac{1}{L} \cdot \frac{1}{J \cdot s} = \frac{1}{L}$$

$$\frac{1}{L} = K_f \frac{L^2}{\hbar^2}$$

$$\hbar q = 2m_e \left(c - \frac{K \cos \theta}{\hbar} \right)$$

↳

$$\frac{\hbar \omega}{c} \rightarrow \omega = \frac{2m_e c^2}{\hbar} -$$

$$\omega = \gamma \epsilon$$

$$1 = \frac{\omega}{c}$$

$$\frac{\hbar \omega}{c}$$

$$\hbar q c \stackrel{!}{=} \frac{\hbar^2}{2m_e} (\vec{k} \cdot \vec{q} + q^2)$$

$$\frac{2m_e c}{\hbar} q = k q \cos \theta + q^2$$

$$q^2 = q \left(\frac{2m_e c}{\hbar} - k \cos \theta \right)$$

non-trivial
solution

$$q = \frac{2m_e c}{\hbar} - k \cos \theta$$

$$|P_{ph}| = \hbar q = 2m_e c - \hbar k \cos \theta \quad \downarrow \cdot c$$

$$\hbar q c = \hbar \omega = \underbrace{2m_e c^2}_{\text{rest energy}} - \hbar c k \cos \theta$$

$$\hbar c k \cos \theta = 2m_e c^2 - E_{ph}$$

$$\approx 2m_e c^2$$

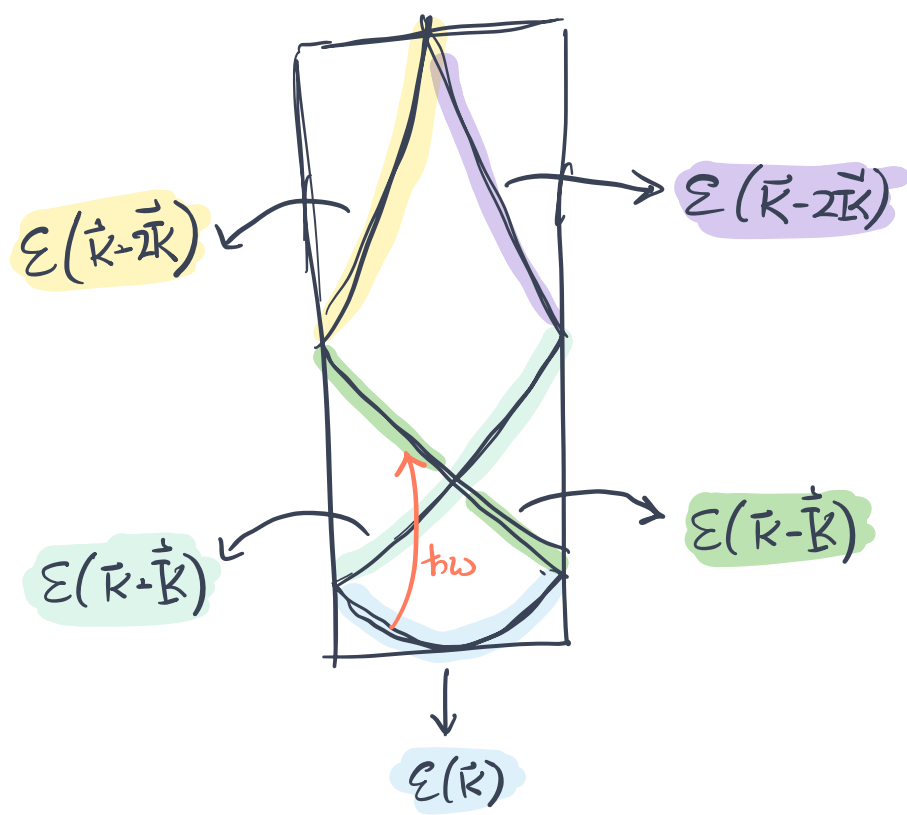
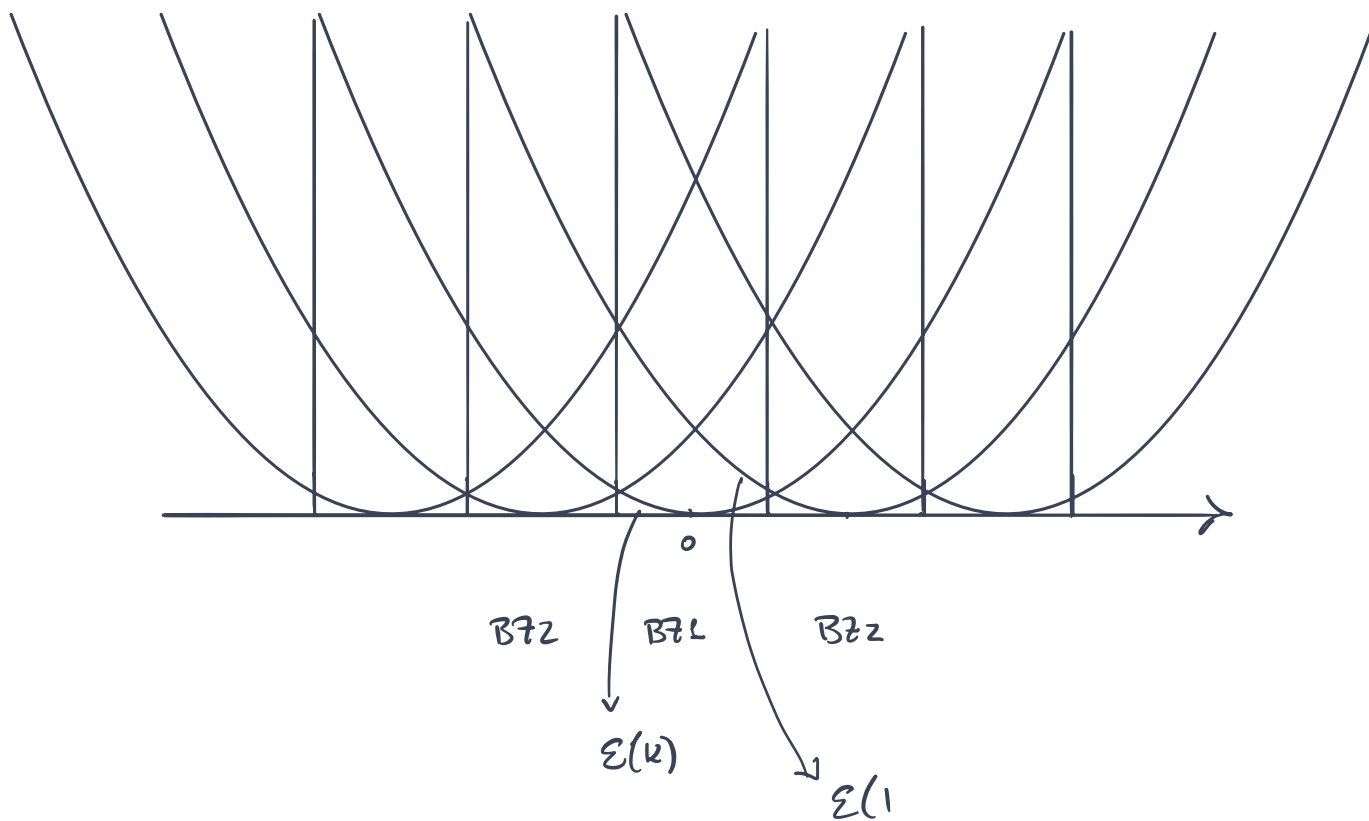
$$k = \frac{2m_e c}{\hbar}$$

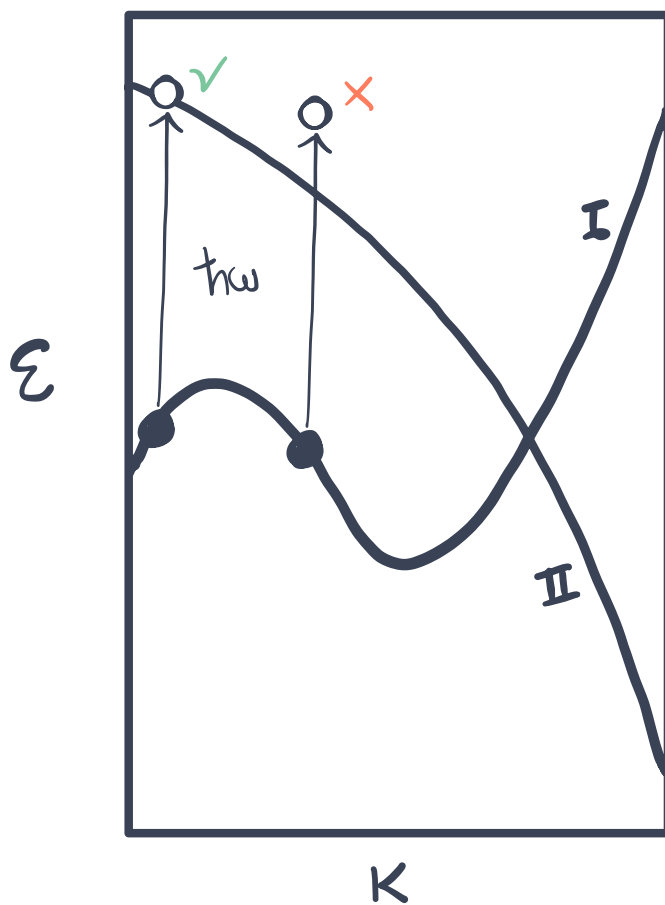
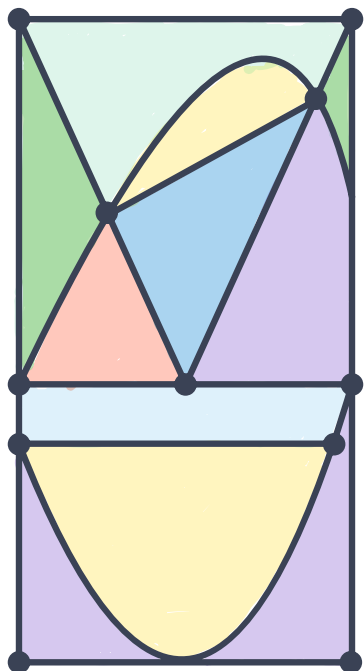
$$\frac{\hbar^2 \left(\frac{2m_e c}{\hbar} \right)^2}{2m_e} = \frac{4m_e^2 c^2}{m_e}, \quad m_e c^2$$

interaction
of free

A bridge between
the classical and quantum
phenomenon of photons
interacting with free electron

photon & free electrons
a bridge between classical
and quantum-mechanical views





$$|\langle \psi_I | H' | \psi_{II} \rangle|^2$$

$$|d| = |\mu_{I,II}|^2 \delta(\vec{k}_{II} - \vec{k}_I - \vec{q})$$

$$\Gamma_{I \rightarrow II} \propto |d|^2 \int f_I(\vec{k}_I) [1 - f_{II}(\vec{k}_I + \vec{q})] d\vec{k}$$

$$\int f_e[\epsilon_I(\vec{k})] f_h[\epsilon_{II}(\vec{k} + \vec{q})] \cdot \delta(\epsilon_{II} - \epsilon_I - \hbar\omega)$$

$$u = \int \frac{d\vec{k}}{4\pi^3} \mathcal{E}(\vec{k}) f(\mathcal{E}(\vec{k}))$$

$$u = 2 \cdot \int \mathcal{E}(\vec{k}) \cdot f[\mathcal{E}(\vec{k})] \frac{d^3k}{(2\pi)^3}$$

Tonight

$$W = \iint_{Bz^2} \frac{2\pi}{\hbar} |\mu_{fi}|^2 \cdot f[E_c(\vec{R}')] \cdot \bar{f}[E_v(\vec{R})] \delta(\vec{R}-\vec{R}') \delta[E_c - E_v - \hbar\omega] \frac{d^3k}{(2\pi)^3} \frac{d^3k'}{(2\pi)^3}$$

$$\frac{d^3k'}{(2\pi)^3} = \frac{1}{(2\pi)^3} \left| \frac{1}{\nabla_{\vec{R}} E_c(\vec{R})} \right| dE_c$$

$$\Downarrow J(\vec{R})$$

$$W = \frac{2\pi}{\hbar} |\mu_{fi}|^2 \cdot \int_{Bz} f[E_v(\vec{R}) + \hbar\omega] \cdot \bar{f}[E_v(\vec{R})] \cdot J(\vec{R}) \cdot \frac{d^3k}{(2\pi)^3}$$



$$\vec{\nabla} \cdot \vec{D} = \rho \quad \vec{\nabla} \times \vec{E} = -\frac{1}{c} \frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \cdot \vec{B} = 0 \quad \vec{\nabla} \times \vec{H}$$

$W(\vec{K}, \vec{K}'; \hbar\omega)$ transition rate from a ^{single} state $\vec{K}$ to a ^{single} state $\vec{K}'$ whose energy difference is $\hbar\omega$

in $E_c(\vec{K})$ ↑ single
 in $E_v(\vec{K})$ ↓

transition rate between a single K state in E_c to all K' states in E_v weighed by the occupancy in each respective state

$$W(K; \hbar\omega) = \int_{BZ} W(\vec{K}, \vec{K}'; \hbar\omega) \bar{f}(E_c(\vec{K}')) \frac{d^3 K'}{(2\pi)^3}$$

$$= \int_{BZ} W(\vec{K}, \vec{K}'; \hbar\omega) \bar{f}(E_c(\vec{K}')) \left| \frac{1}{\nabla_{\vec{K}} E_c} \right| dE_c$$

$$W(\vec{K}, \vec{K}'; \hbar\omega) \propto |\mu_{cv}|^2 \langle \vec{K}' | \vec{K} - \vec{q} \rangle \cdot \langle E_c | E_v + \hbar\omega \rangle$$

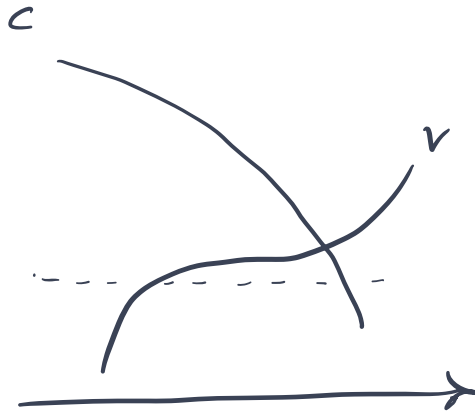
[

$$W(\hbar\omega) = \int_{BZ} W(\vec{K}, \vec{K}'; \hbar\omega) \cdot \bar{f}(E_{\#}(\vec{K}')) \frac{d^3 K'}{(2\pi)^3}$$

$$\Omega_{vc}(\vec{R}) = 0$$

$$\delta[E_v(\vec{R}') - E_c(\vec{R}) + \hbar\omega] = \sum_{\vec{R}_i'} \frac{1}{|\vec{\nabla}_{\vec{R}_i'} \Delta E|_{\vec{R}_i'}} \cdot \delta(\vec{R} - \vec{R}_i')$$

s.t. $\Delta E(\vec{R}_i') = 0$



- o FGR is derived by considering the perturbation of a harmonic field two

$$\delta[f(x)] = \sum_{\substack{x_i \\ \text{s.t. } f(x_i)=0}} \frac{1}{|f'(x_i)|} \delta(x-x_i)$$

$$\delta[f(\vec{x})] = \sum_{\vec{x}_i: f(\vec{x}_i)=0} \frac{1}{|\vec{\nabla} f(\vec{x}_i)|} \delta(\vec{x}-\vec{x}_i)$$

$$\int_{BZ} \delta[E_v(\vec{k}') - E_c(\vec{k}) + \hbar\omega] d^3k' = \int_S \frac{1}{|\vec{\nabla}_{\vec{k}'} E_v(\vec{k}')|} dS$$

$$W \propto \int_{BZ} \int_{BZ} f[E_c(\vec{R})] \bar{f}[E_v(\vec{R}')] \delta[E_v(\vec{R}') - E_c(\vec{R}) - \hbar\omega] \frac{d^3 \vec{K}'}{(2\pi)^3} \frac{d^3 \vec{K}}{(2\pi)^3}$$

$$= \int_{BZ} f[E_c(\vec{R})] \left\{ \int_{BZ} \bar{f}[E_v(\vec{R}')] \delta[E_v(\vec{R}') - E_c(\vec{R}) - \hbar\omega] \frac{d^3 \vec{K}'}{(2\pi)^3} \right\} \frac{d^3 \vec{K}}{(2\pi)^3}$$

$$\int_{BZ} \frac{\bar{f}[E_v(\vec{R}')] }{|\vec{\nabla}_{\vec{K}'} E_v(\vec{R}_0)|} \delta(\vec{R}' - \vec{R}_0) d^3 \vec{K}$$

where $\vec{R}_0$ satisfies

$$E_v(\vec{R}_0) = E_c(\vec{R}) - \hbar\omega$$

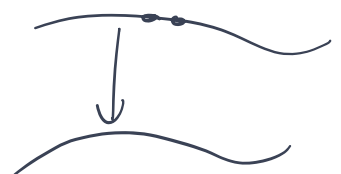
$$= \frac{\bar{f}[E_v(\vec{R}_0)]}{|\vec{\nabla}_{\vec{K}'} E_v(\vec{R}_0)|} = \frac{\bar{f}[E_c(\vec{R}) - \hbar\omega]}{|\vec{\nabla}_{\vec{K}'} E_v(\vec{R}_0)|}$$

$$= \frac{1}{(2\pi)^3} \int_{BZ} f[E_c(\vec{R})] \bar{f}[E_c(\vec{R}) - \hbar\omega] \frac{d^3 \vec{K}}{|\vec{\nabla}_{\vec{K}'} E_v(\vec{R}_0)|}$$

Since $\hbar\omega \ll E_c, E_v \Rightarrow \vec{R} = \vec{R}' = \vec{R}_0$

$$\frac{1}{(2\pi)^3} \int_{BZ} f[E_c(\vec{R})] \{1 - f[E_c(\vec{R}) - \hbar\omega]\} \frac{d\vec{R}}{|\vec{\nabla} E_v(\vec{R})|}$$

$$\langle \varepsilon_i | H' | \varepsilon_f \rangle$$



$$\Gamma_{i \rightarrow f}(\hbar\omega) = \frac{2\pi}{\hbar} |\langle i | H' | f \rangle|^2 \cdot \delta(\epsilon_f - \epsilon_i + \hbar\omega)$$

$$\Gamma_i(\hbar\omega) = \sum_{\epsilon_f} \frac{2\pi}{\hbar} |\mu_{if}|^2 \cdot \delta(\epsilon_f - \epsilon_i + \hbar\omega)$$

$\Gamma_{i \rightarrow f}$

$$V \int \frac{2\pi}{\hbar} |\mu_{if}|^2 \delta(\epsilon_f - \epsilon_i + \hbar\omega) \underbrace{[1 - f(\epsilon_f)] \cdot \rho(\epsilon_f) d\epsilon_f}_{\text{blue underline}}$$

$$\Gamma(\hbar\omega) = V \int \Gamma_i(\hbar\omega) f(\epsilon_i) \rho(\epsilon_i) d\epsilon_i$$

$$= \frac{2\pi V^2}{\hbar} \iint |\mu_{if}|^2 \rho_{\mathcal{J}}(\epsilon_i, \epsilon_f) \delta(\epsilon_f - \epsilon_i + \hbar\omega) d\epsilon_i d\epsilon_f$$

$$f(\epsilon_i) \rho(\epsilon_i) \cdot [1 - f(\epsilon_f)] \rho(\epsilon_f) \equiv \rho_{\mathcal{J}}(\epsilon_i, \epsilon_f)$$

$$= \frac{2\pi V^2}{\hbar} \int |\mu(\epsilon_i, \epsilon_i - \hbar\omega)|^2 \rho_{\mathcal{J}}(\epsilon_i, \epsilon_i - \hbar\omega) d\epsilon_i //$$

transition
probability
rate

$$\left| \delta_{\vec{k}_f + \vec{q}, \vec{k}_i} \right| \quad \mu.$$

$$\Gamma_{i \rightarrow f}(\hbar\omega) = \left[\frac{2\pi}{\hbar} |\langle i | H' | f \rangle|^2 \cdot \delta(\epsilon(\vec{k}_f) - \epsilon(\vec{k}_i) + \hbar\omega) \right]$$

$$\bar{f}(\vec{k}_f)$$



$$\Gamma_i(\hbar\omega) = V \int \Gamma_{i \rightarrow f}(\hbar\omega) [1 - f(\vec{k}_f)] \frac{d^3 \vec{k}_f}{(2\pi)^3}$$

$$\Gamma(\hbar\omega) = V \int \Gamma_i(\hbar\omega) f(\vec{k}_i) \frac{d^3 \vec{k}_i}{(2\pi)^3}$$

$$= \frac{2\pi V^2}{\hbar} \int |\mu(\vec{k}_i, \vec{k}_f)|^2 f(\vec{k}_i) \bar{f}(\vec{k}_f) \delta(\epsilon(\vec{k}_f) - \epsilon(\vec{k}_i) + \hbar\omega) \frac{d^3 \vec{k}_i}{(2\pi)^3} \frac{d^3 \vec{k}_f}{(2\pi)^3}$$

$$\delta(f(x)) = \sum_{\substack{x_0 \\ \text{s.t. } f(x_0)=0}} \frac{\delta(x_0)}{|f'(x_0)|}$$

$$f(x) = u.$$

$$\int \delta(\overbrace{f(x)}^u) dx = \int \delta(u) \frac{du}{f'(x)}$$

$$\uparrow$$

$$du = f'(x) dx$$

$$\int h(x) \delta(f(x)) dx =$$

$$f(x) = u \rightarrow x = f^{-1}(u)$$

$$f'(x) dx = du$$

$$\uparrow$$

$$\int h[f^{-1}(u)] \delta(u) \frac{du}{f'[f^{-1}(u)]} = \frac{h[f^{-1}(0)]}{f'[f^{-1}(0)]}$$

$$\exists \{x_0\}_n : \text{s.t. } f(x_0) = 0$$

$$f \rightarrow \sum_{(k_f)} \mathcal{E}(k_f) - \mathcal{E}(k_i) + \hbar \omega$$

$$\sum_{x_0} \frac{h(x_0)}{f'(x_0)}$$

$$k_0 \text{ st } \mathcal{E}(k_0) = \mathcal{E}(k_i) + \hbar \omega$$

$$= \frac{2\pi V^2}{\hbar} \int |\mu(k_i, \underline{k_f})|^2 \underline{f}(k_i) \bar{\underline{f}}(\underline{k_f}) \delta(\mathcal{E}(k_f) - \mathcal{E}(k_i) + \hbar \omega) \frac{d^3 k_i}{(2\pi)^3} \frac{d^3 k_f}{(2\pi)^3}$$

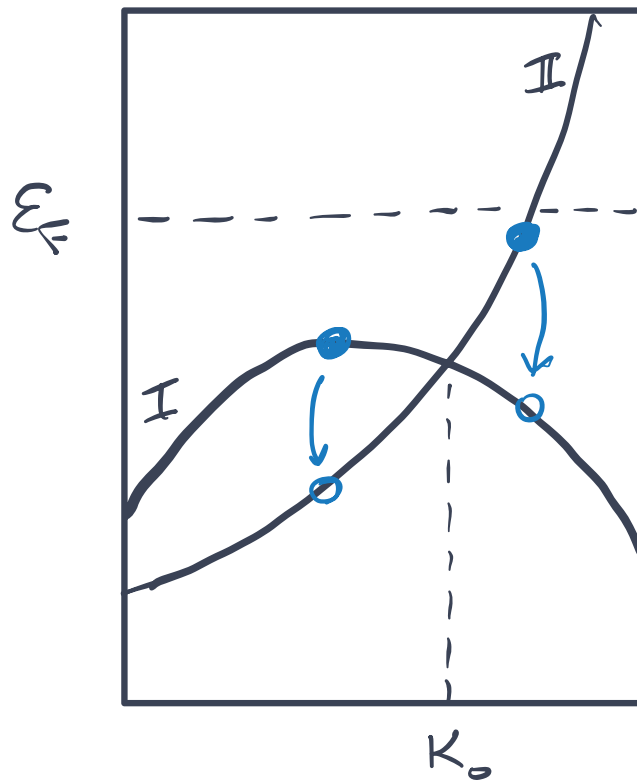
$$\frac{2\pi V^2}{\hbar} \int f(k_i) \left[\int |\mu(k_i, k_f)|^2 \bar{f}(k_f) \cdot \delta(\mathcal{E}(k_f) - \overbrace{\mathcal{E}(k_i)}^E + \hbar \omega) \frac{d^3 k_f}{(2\pi)^3} \right] \frac{d^3 k_i}{(2\pi)^3}$$

$$h(k_f) = \frac{|\mu(\vec{k}_i, \vec{k}_f)|^2 \bar{f}(\vec{k}_f)}{2\pi}$$

$$\int h(\vec{k}_f) \cdot \delta(\mathcal{E}(\vec{k}_f) - E) d^3 k_f$$

$$\Sigma(K_{f,i}) = u_x$$

$$\frac{d\varepsilon}{dK_{f,j}} dK_{f,j} = du_j \rightarrow dK_{f,j} = \left| \frac{d\varepsilon}{dK_{f,j}} \right|^{-1} du_j$$



$$\Gamma_0 = \Gamma_{I \rightarrow II}(K) + \Gamma_{II \rightarrow I}$$

$$f: \mathbb{R}^n \rightarrow \mathbb{R}^m$$

$$J_f = \begin{bmatrix} \frac{\partial \vec{f}}{\partial x_1} & \dots & \frac{\partial \vec{f}}{\partial x_n} \end{bmatrix}$$

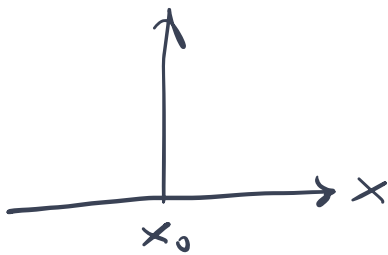
$$= \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \dots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}$$

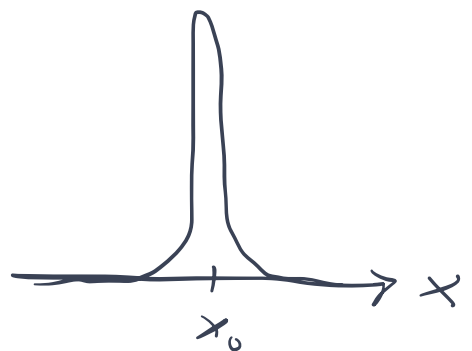
$$\vec{f}(\vec{x}) = \vec{f}(\vec{x} - \vec{x}_0) + \frac{\partial \vec{f}_i}{\partial x_k} (x_k - x_k^0)$$

$$f_i(x_k) = f_i(x_k - x_k^0) + \frac{\partial f_i}{\partial x_k} (x_k - x_k^0)$$

$$= f_i^0(x_k - x_k^0) + J_{ik}^0 (x_k - x_k^0)$$

Dirac-delta Gaussian Approximation

$$\delta(x-x_0) =$$


$$G(x-x_0) =$$


$$= \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\left(\frac{x-x_0}{\sqrt{2}\sigma}\right)^2\right]$$

$$G(\underbrace{\varepsilon_1(k_f) - \varepsilon_2(k_i)}_{\Delta(k_i)} + \hbar\omega)$$

$$= \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\left(\frac{\varepsilon_1(k_f) - \Delta(k_i)}{\sqrt{2}\sigma}\right)^2\right]$$

$$= \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\left(\frac{\varepsilon_1^2(k_f) + \Delta^2(k_i) - \varepsilon_1(k_f)\Delta(k_i)}{2\sigma^2}\right)\right]$$

$$= \frac{1}{\sigma \sqrt{2\pi}} e^{-\left(\frac{\Sigma_1(K_f)}{\sqrt{2}\sigma}\right)^2} e^{-\left(\frac{\Sigma_2(K_i)}{\sqrt{2}\sigma}\right)^2} e^{\frac{\Sigma_1(K_f) \cdot \Delta(K_i)}{2\sigma^2}}$$

$$\left[\begin{array}{cc} \text{O}^{e_1} & \text{O}^{e_2} \\ \text{N} \text{---} \text{O}^{\text{P}} \text{---} \text{O}^{\text{P}} \text{---} \text{N} \end{array} \right] : H_{H_0} = K_{e_1} + K_{e_2} + K_n$$

$$+ V_{e_1 n} + V_{e_2 n} + V_{e_1 e_2}$$



$$H_{H_2} = K_{e_1} + K_{e_2} + K_{P_1} + K_{P_2}$$

$$+ V_{e_1 P_1} + V_{e_1 P_2} + V_{e_2 P_1} + V_{e_2 P_2}$$

$$+ V_{e_1 e_2} + V_{P_1 P_2}$$

$$R = \frac{2\pi}{\hbar} |H| \rho$$

$$E: |e\rangle \otimes |0\rangle_1, |0\rangle_2, \dots$$

$$G: |g\rangle \otimes \begin{array}{l} |1\rangle_1, |0\rangle_2, |0\rangle_3, \dots \\ \text{or} \\ |0\rangle_1, |1\rangle_2, |0\rangle_3, \dots \end{array} \left. \vphantom{\begin{array}{l} |1\rangle_1, |0\rangle_2, |0\rangle_3, \dots \\ \text{or} \\ |0\rangle_1, |1\rangle_2, |0\rangle_3, \dots \end{array}} \right\} \begin{array}{l} \text{integrate} \\ \text{over all} \\ \text{possibilities} \end{array}$$

$$|H_{GE}|^2 = e^2 \epsilon_1^2 (\vec{r}_{ge} \cdot \hat{e})^2 |n+1\rangle$$



How do you get to this
term

$$|e\rangle \langle g|$$

Quantization of the Radiation Field

$$\vec{A}(\vec{r}, t) = \hat{e} A_0 e^{i(\vec{R} \cdot \vec{r} - \omega t)} + c.c$$

$$\vec{E}(\vec{r}, t) = - \frac{\partial \vec{A}}{\partial t}$$

total energy

Potential energy
quadratic in amplitude
hints at a harmonic
oscillator

volume
of space

$$U = V \cdot \frac{\epsilon_0}{2} \omega^2 |A_0|^2$$

Define Q, P

$$A = \sqrt{\frac{1}{\epsilon_0 V}} \frac{1}{\omega} (\omega Q + iP)$$

$$U = \frac{1}{2} (\omega^2 Q^2 + P^2)$$

$$[Q, P] = i\hbar$$

Quantum
leap

↓
let us describe
this as a QHO.

$$a = \frac{1}{\sqrt{2\hbar\omega}} (\omega Q + iP)$$

$$a^\dagger = \frac{1}{\sqrt{2\hbar\omega}} (\omega Q - iP)$$

$$[a, a^\dagger] = 1$$

$$U \rightarrow H = \hbar\omega [a^\dagger a + \frac{1}{2}]$$

$$\langle n-1 | a | n \rangle = \sqrt{n}$$

$$\langle n-1 | a^\dagger | n \rangle = \sqrt{n+1}$$

$$\omega = c|q| = cq$$

$$E = i \sqrt{\frac{\hbar\omega}{2\epsilon_0 V}} \left[\hat{\mathbf{a}} e^{i(\mathbf{R} \cdot \vec{q} - \omega t)} - c.c \right]$$

$$E_q^\mu = i \vec{E}_{\perp,q} \left[e_\mu a_{\vec{q},\mu} e^{i\vec{q} \cdot \vec{r}} - e_\mu^\dagger a_{\vec{q},\mu}^\dagger e^{-i\vec{q} \cdot \vec{r}} \right]$$



electric field
of a single photon

$$\hat{e}_\mu = \hat{e}_{(\pm 1)} = \frac{\mp 1}{\sqrt{2}} (\hat{e}_x \pm i \hat{e}_y)$$

conveniently
locate
the quantum
emitter at
 $\vec{r}_0$ such that
 $e^{i\vec{q} \cdot \vec{r}_0} = 1$

$$\hat{e}_{(1)} = -\frac{1}{\sqrt{2}} (\hat{e}_x + i \hat{e}_y) \quad \hat{e}_{(-1)} = \frac{1}{\sqrt{2}} (\hat{e}_x - i \hat{e}_y) = -\hat{e}_{(1)}^\dagger$$

$$\hat{e}_{(1)}^\dagger = -\frac{1}{\sqrt{2}} (\hat{e}_x - i \hat{e}_y) \quad \hat{e}_{(-1)}^\dagger = \frac{1}{\sqrt{2}} (\hat{e}_x + i \hat{e}_y) = -\hat{e}_{(1)}$$

$$E_{q,\mu} = i E_{1,\vec{q}} \left[\hat{e}_\mu a_{\vec{q},\mu} - \hat{e}_\mu^\dagger a_{\vec{q},\mu}^\dagger \right]$$

lets neglect μ DOF

$$E_{\vec{q}} = i E_{1,\vec{q}} \left[\hat{e} a_{\vec{q}} - \hat{e}^\dagger a_{\vec{q}}^\dagger \right]$$

$$H_{int}^q = -\vec{d} \cdot \vec{E}_q = e \vec{r} \cdot \vec{E}_{\vec{q}}$$

$$= i e E_{1,q} \vec{r} \cdot \left[\hat{e} a_{\vec{q}} - \hat{e}^\dagger a_{\vec{q}}^\dagger \right]$$

$$|i\rangle = |e\rangle \otimes |n\rangle$$

$$|f\rangle = |g\rangle \otimes |n+1\rangle$$

more general initial
and final states

$$|i\rangle = |a\rangle \otimes |n\rangle$$

$$|f\rangle = |b\rangle \otimes |n\rangle$$

$$\langle f | H_{int} | i \rangle = ieE_{1,q} (\vec{r}_{a,b} \cdot \hat{e})$$

$$\langle b, n' | \vec{r} (a + a^\dagger) | a, n \rangle$$

$$= \langle b | \vec{r} | a \rangle \langle n' | a + a^\dagger | n \rangle$$

$$= \vec{r}_{ab} \left(\sqrt{n} \langle n' | n-1 \rangle + \sqrt{n+1} \langle n' | n+1 \rangle \right)$$



absorption

$n' \rightarrow n-1$

a photon is
subtracted from
the field.



emission

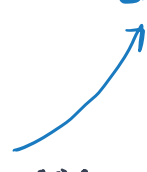
$n' \rightarrow n+1$

a photon added
to the field

Spontaneous Emission

$$R = 2\pi \frac{|H_{EG}|^2}{\hbar^2} \rho(\omega)$$

$$E = |e\rangle \otimes |0\rangle_1 |0\rangle_2 |0\rangle_3 \dots$$

EM mode 

$$G = |g\rangle \otimes |1\rangle_1 |0\rangle_2 |0\rangle_3 \dots$$

or

$$|0\rangle_1 |1\rangle_2 |0\rangle_3 \dots$$

the photon emitted can theoretically appear in any of the modes, so one should integrate over all possibilities

(A delta should appear as a constraint allowing only emission to modes that conserve energy given a $|e\rangle \rightarrow |g\rangle$ transition.

we start with 0
photons

$$|H_{GE}|^2 = e^2 \cdot E_1^2 \cdot (\vec{r}_{ge} \cdot \hat{\theta})^2 \cdot (n+1)$$



$$= e^2 \frac{\hbar \omega}{2 \epsilon_0 V} (\vec{r}_{ge} \cdot \hat{e})$$

the remaining 1 can
be thought of as a
coupling due to vacuum
fluctuations

$$\frac{d\Gamma}{d\Omega} =$$

freq of
emitted
photon

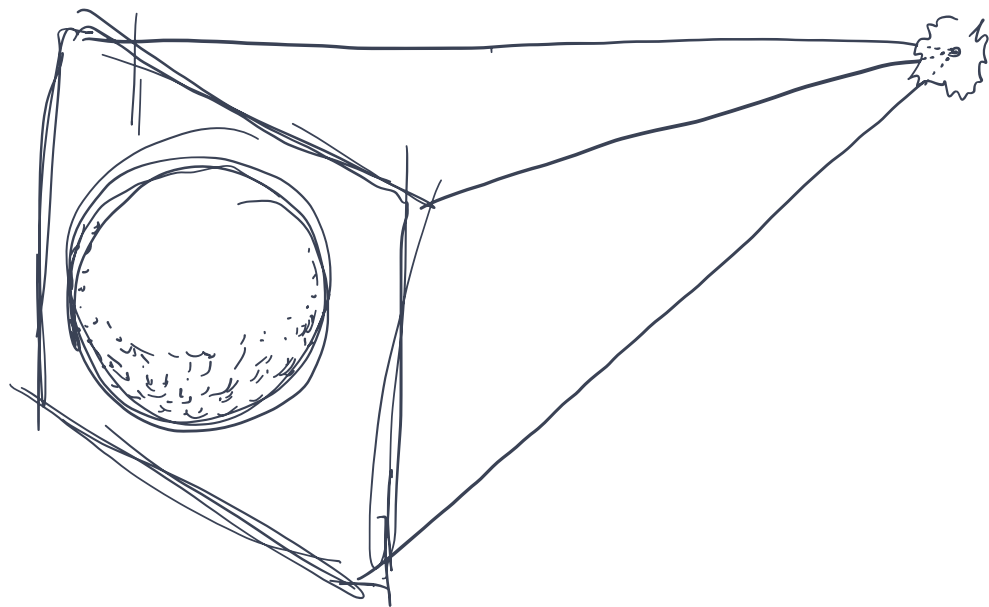
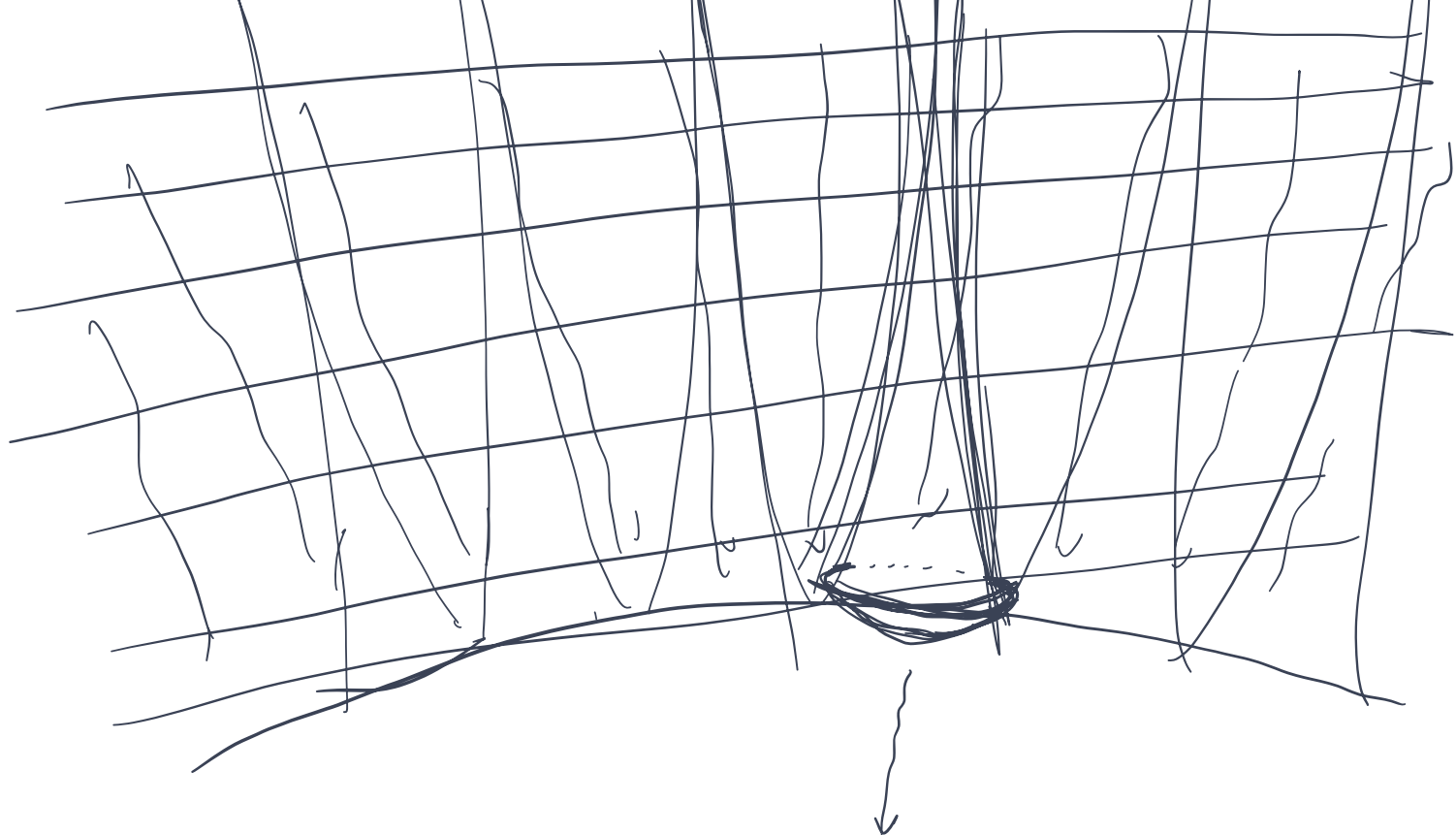
$$\delta(\omega_K - \omega_0)$$

$\omega_0 - \omega_f$

$$\delta[\hbar\omega - \epsilon_c + \epsilon_v]$$

$$\delta[\epsilon_v - \epsilon_c + \hbar\omega]$$

$$\gamma = \frac{2\omega}{\hbar}$$



$$P = I_{\odot}(d_{\odot \rightarrow \oplus}) \cdot S_{\oplus}$$

$$|\langle f | v | i \rangle|^2 = \langle i | \hat{\mu} \cdot \hat{E} | f \rangle \langle f | \hat{\mu} \cdot \hat{E} | i \rangle$$

$$\left\{ \begin{array}{l} \hat{E} = \sum_q \left[E_q^+ a_q(t) + E_q^- a_q^\dagger(t) \right] \\ a_q(t) = a_q(0) e^{-i\omega_q t} \\ a_q^\dagger(t) = a_q^\dagger(0) e^{i\omega_q t} \end{array} \right\}$$

$$\begin{aligned} \hat{\mu} \cdot \hat{E} = \sum_q \mu \cdot & \left[E_q^+ \hat{\tau}^+ \hat{a}_q(t) \right. \\ & + E_q^- \hat{\tau}^- \hat{a}_q^\dagger(t) \\ & + E_q^+ \hat{\tau}^- \hat{a}_q(t) \\ & \left. + E_q^- \hat{\tau}^+ \hat{a}_q^\dagger(t) \right] \end{aligned}$$

$$|i\rangle = |e, \{0\}\rangle \quad \text{excited state + EM vacuum}$$

$$|f\rangle = |g, \{1_q\}\rangle \quad \text{ground state + 1 EM mode } q.$$

$$\langle f | \hat{\mu} \cdot \hat{E} | i \rangle = \left\langle g, \{1_q\} \right| \sum_q \mu \cdot \left[\cancel{E_q^+} \hat{\tau}^+ \hat{a}_q(t) + E_q^- \hat{\tau} \hat{a}_q^\dagger(t) + E_q^+ \hat{\tau} \cancel{a_q(t)} + E_q^- \hat{\tau}^+ \cancel{\hat{a}_q^\dagger(t)} \right] \left| \underline{e}, \{0\} \right\rangle$$

$$= \langle g, \{1_q\} | E_q^- \hat{\tau} \hat{a}_q^\dagger(t) | e, \{0\} \rangle$$

$$= E_q^- \langle g, \{1_q\} | g, \{1_q\} \rangle = E_q^-$$

$$\langle i | \hat{\mu} \cdot \hat{E} | f \rangle = \left\langle e, \{0\} \right| \sum_q \mu \cdot \left[E_q^+ \hat{\tau}^+ \hat{a}_q(t) + \cancel{E_q^-} \hat{\tau} \cancel{\hat{a}_q^\dagger(t)} + \cancel{E_q^+} \hat{\tau} \hat{a}_q(t) + E_q^- \hat{\tau}^+ \cancel{\hat{a}_q^\dagger(t)} \right] \left| \underline{g}, \{1_q\} \right\rangle$$

$$= \langle e, \{0\} | E_q^+ \hat{\tau}^+ \hat{a}_q(t) | g, \{1_q\} \rangle$$

$$= \bar{E}_q^+ \langle e, \{0\} | e, \{0\} \rangle = \bar{E}_q^+$$

$$|\langle f | v | i \rangle|^2 = \sum_q [(\vec{\mu} \cdot \vec{E}_q^+) (\vec{E}_q^- \cdot \vec{\mu})]$$

$$\left\{ \begin{array}{l} \vec{E}_q^+ = \sqrt{\frac{\hbar \omega_q}{2\epsilon_0}} \vec{u}_q \quad \vec{E}_q^- = \sqrt{\frac{\hbar \omega_q}{2\epsilon_0}} \vec{u}_q^* \\ \vec{\mu} = |\mu| \vec{n} \end{array} \right\}$$

$$|\langle f | v | i \rangle|^2 = \mu^2 \sum_q \frac{\hbar \omega_q}{2\epsilon_0} [(\vec{n} \cdot \vec{u}_q)(\vec{n} \cdot \vec{u}_q^*)] \delta(\omega$$

$$\delta(\omega_q - \omega_0)$$

$$\omega_0 = \omega_e - \omega_g$$

$$\delta(\hbar \omega_q - \epsilon_e - \epsilon_g)$$

$$\delta(\underline{\epsilon_f - \epsilon_i} + \hbar \omega_q)$$

$$I = \frac{J}{A s}$$

$$u = \frac{I}{c} = \frac{J}{v}$$